



ACADEMIC YEAR/ANNEE ACADEMIQUE 2025/2026

DOMAIN: Computer Science	COUR: Programming III and Advanced Databases
SPECIALTE :Software Engineering	NIVEAU :2
SESSION d'EXAMEN: PREMIER SEMESTRE	ENCADREUR:Mr Djakba Faustin
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PART 1: General Knowledge in Programming III and Advanced Databases

HTML (Questions 1-15)

- What does HTML stand for?**
 - a) Hyper Text Markup Language
 - b) High Tech Modern Language
 - c) Home Tool Markup Language
 - d) Hyperlinks and Text Markup Language
- Which HTML tag is used to define the main content of a document?**
 - a) <body>
 - b) <main>
 - c) <content>
 - d) <section>
- Which attribute is used to provide a unique identifier for an HTML element?**
 - a) class
 - b) name
 - c) id
 - d) style
- What is the correct HTML element for inserting a line break?**
 - a) <lb>
 - b) <break>
 - c)

 - d) <newline>
- Which HTML tag is semantic and represents independent, self-contained content?**
 - a) <div>

- b)
- c) <article>
- d) <section>

6. How do you create a hyperlink in HTML?

- a) <link url="http://example.com">
- b)
- c) <hyperlink>http://example.com</hyperlink>
- d)

7. Which HTML element is used to define an internal style sheet?

- a) <css>
- b) <script>
- c) <style>
- d) <design>

8. What is the correct HTML for adding a background color?

- a) <body bg="yellow">
- b) <body style="background-color:yellow;">
- c) <background>yellow</background>
- d) <body color="yellow">

9. Which input type is used for email validation in HTML5?

- a) text
- b) mail
- c) email
- d) validate

10. What does the alt attribute in an tag specify?

- a) Image alignment
- b) Alternative text if image cannot be displayed
- c) Image source URL
- d) Image size

11. Which HTML5 element is used for navigation links?

- a) <nav>
- b) <navigation>
- c) <links>
- d) <menu>

12. How do you define an HTML form?

- a) <form>
- b) <input type="form">
- c) <fieldset>
- d) <formfield>

13. Which attribute makes an input field required in HTML5?

- a) validate
- b) required
- c) mandatory
- d) check

14. What is the purpose of the <meta> tag?

- a) Display metadata on the page
- b) Provide metadata about the HTML document
- c) Create meta-navigation
- d) Define meta-styles

15. Which HTML element is used to embed JavaScript?

- a) <js>
- b) <javascript>
- c) <script>
- d) <code>

CSS (Questions 16-30)

16. What does CSS stand for?

- a) Creative Style Sheets
- b) Computer Style Sheets
- c) Cascading Style Sheets
- d) Colorful Style Sheets

17. Which CSS property controls the text size?

- a) font-style
- b) text-size
- c) font-size
- d) text-style

18. How do you select an element with id "header"?

- a) .header
- b) #header

- c) header
- d) *header

19. Which CSS property is used to change the background color?

- a) color
- b) bgcolor
- c) background-color
- d) background

20. What is the correct CSS syntax?

- a) body: color=black;
- b) {body: color=black;}
- c) body {color: black;}
- d) body {color: black}

21. Which property is used to create flexible box layouts?

- a) grid
- b) flex
- c) display: flexbox
- d) layout: flexible

22. How do you make a list without markers?

- a) list: none;
- b) list-style-type: none;
- c) marker: none;
- d) bullet: none;

23. Which CSS property controls element visibility?

- a) display
- b) visibility
- c) opacity
- d) show

24. What does the box-sizing: border-box; property do?

- a) Adds border to element
- b) Includes padding and border in element's width/height
- c) Removes box shadows
- d) Creates a border-radius

25. Which unit is relative to the font-size of the root element?

- a) em

- b) px
- c) rem
- d) %

26. How do you apply CSS to all <p> elements inside a <div>?

- a) div + p
- b) div > p
- c) div.p
- d) div p

27. Which property adds space outside an element's border?

- a) padding
- b) margin
- c) spacing
- d) border-spacing

28. What is the default position value in CSS?

- a) relative
- b) absolute
- c) fixed
- d) static

29. Which CSS3 property adds shadow to text?

- a) box-shadow
- b) text-shadow
- c) font-shadow
- d) shadow-text

30. How do you create a responsive design that adapts to screen size?

- a) fluid layouts
- b) media queries
- c) flexible units only
- d) JavaScript resizing

JavaScript (Questions 31-50)

31. Inside which HTML element do we put JavaScript?

- a) <js>
- b) <scripting>
- c) <script>

- d) <javascript>

32. What is the correct JavaScript syntax to write "Hello World"?

- a) echo "Hello World";
- b) console.log("Hello World");
- c) print("Hello World");
- d) document.write("Hello World");

33. Which operator is used for strict equality comparison?

- a) =
- b) ==
- c) ===
- d) ===

34. How do you declare a JavaScript variable?

- a) variable myVar;
- b) v myVar;
- c) let myVar;
- d) var myVar; (also correct but let is ES6+ preferred)

35. Which method adds an element to the end of an array?

- a) append()
- b) push()
- c) add()
- d) insert()

36. What does DOM stand for?

- a) Document Object Model
- b) Data Object Model
- c) Document Orientation Model
- d) Digital Object Management

37. Which event occurs when a user clicks on an HTML element?

- a) onchange
- b) onmouseleave
- c) onclick
- d) onpress

38. How do you select an element with id "demo" in JavaScript?

- a) document.getElement("demo")
- b) document.getElementById("demo")

- c) document.getElementById("demo")
- d) document.select("demo")

39. Which function is used to parse a JSON string?

- a) JSON.parse()
- b) JSON.decode()
- c) parseJSON()
- d) JSON.toString()

40. What is the result of typeof [] in JavaScript?

- a) "array"
- b) "object"
- c) "list"
- d) "undefined"

41. Which keyword is used to define a function in ES6+?

- a) func
- b) function
- c) def
- d) arrow

42. What does event.preventDefault() do?

- a) Stops event propagation
- b) Prevents default browser behavior
- c) Cancels the event entirely
- d) Delays event execution

43. Which method converts a JSON object to a string?

- a) JSON.toString()
- b) JSON.stringify()
- c) JSON.encode()
- d) toStringJSON()

44. How do you write an IF statement in JavaScript?

- a) if i = 5 then
- b) if (i == 5)
- c) if i == 5 then
- d) if i = 5

45. Which loop is guaranteed to execute at least once?

- a) for

- b) while
- c) do...while
- d) foreach

46. What is a closure in JavaScript?

- a) A function with no parameters
- b) A function that has access to outer scope variables
- c) A closed loop structure
- d) A terminated function

47. Which method adds a new element to the beginning of an array?

- a) push()
- b) unshift()
- c) prepend()
- d) start()

48. What does async/await handle in JavaScript?

- a) Synchronous code only
- b) Promises and asynchronous operations
- c) Error handling only
- d) DOM manipulation

49. Which property returns the number of characters in a string?

- a) size
- b) count
- c) length
- d) size()

50. How do you call a function named "myFunction"?

- a) call myFunction()
- b) myFunction()
- c) run myFunction
- d) execute myFunction()

PHP (Questions 51-70)

51. What does PHP stand for?

- a) Private Home Page
- b) Personal Hypertext Processor
- c) PHP: Hypertext Preprocessor

- d) Preprocessed Home Page

52. Which symbol is used to start a PHP block?

- a) <?
- b) <?php
- c) <php>
- d) <?php?>

53. How do you create a variable in PHP?

- a) \$myVar = "value";
- b) var myVar = "value";
- c) variable myVar = "value";
- d) myVar = "value";

54. Which superglobal contains form data sent via POST?

- a) \$_GET
- b) \$_POST
- c) \$_REQUEST
- d) \$_FORM

55. What is the correct way to include a file in PHP?

- a) <!-- include file.php -->
- b) <? include 'file.php'; ?>
- c) <include src="file.php">
- d) import file.php;

56. Which function is used to connect to a MySQL database in modern PHP?

- a) mysql_connect()
- b) mysqli_connect()
- c) pdo_connect()
- d) db_connect()

57. How do you output text in PHP?

- a) echo "Hello";
- b) print("Hello"); (also valid)
- c) console.log("Hello");
- d) write("Hello");

58. Which PHP statement is used to stop execution and show an error?

- a) break
- b) exit or die()

- c) stop
- d) end

59. What does the === operator check in PHP?

- a) Value only
- b) Value and type
- c) Reference only
- d) Object identity

60. Which function sanitizes user input to prevent XSS?

- a) strip_tags()
- b) htmlspecialchars()
- c) clean_input()
- d) sanitize()

61. How do you start a session in PHP?

- a) session_start()
- b) start_session()
- c) session_begin()
- d) init_session()

62. Which array type in PHP has named keys?

- a) Indexed array
- b) Associative array
- c) Multidimensional array
- d) Numeric array

63. What is the purpose of prepared statements in PHP/MySQL?

- a) Speed up queries
- b) Prevent SQL injection
- c) Format output
- d) Cache results

64. Which function checks if a variable is set in PHP?

- a) isset()
- b) exists()
- c) defined()
- d) check()

65. How do you redirect a user to another page in PHP?

- a) redirect("page.php");

- b) header("Location: page.php");
- c) goto("page.php");
- d) location("page.php");

66. Which PHP extension is recommended for database operations?

- a) mysql
- b) mysqli
- c) pdo ☐ (both b and c are valid; PDO is more flexible)
- d) dbi

67. What does \$_SERVER['PHP_SELF'] return?

- a) Server IP address
- b) Current script filename/path
- c) User agent string
- d) Request method

68. Which function is used to hash passwords securely in PHP?

- a) md5()
- b) sha1()
- c) password_hash()
- d) crypt()

69. How do you comment a single line in PHP?

- a) <!-- comment -->
- b) // comment
- c) /* comment */
- d) # comment (also valid)

70. What is the default method for HTML forms?

- a) POST
- b) GET
- c) PUT
- d) REQUEST

SQL/MySQL (Questions 71-90)

71. What does SQL stand for?

- a) Structured Query Language
- b) Simple Query Language
- c) Standard Question Language
- d) System Query Logic

72. Which SQL statement is used to extract data from a database?

- a) GET
- b) OPEN
- c) SELECT
- d) EXTRACT

73. Which clause is used to filter records in SQL?

- a) FILTER
- b) WHERE
- c) HAVING
- d) CONDITION

74. What does CRUD stand for in database operations?

- a) Create, Read, Update, Delete
- b) Copy, Remove, Upload, Download
- c) Connect, Request, Use, Disconnect
- d) Compile, Run, Update, Debug

75. Which SQL keyword is used to sort the result-set?

- a) SORT BY
- b) ORDER BY
- c) ARRANGE BY
- d) GROUP BY

76. What is the purpose of a PRIMARY KEY?

- a) To create relationships
- b) To uniquely identify each record
- c) To encrypt data
- d) To index all columns

77. Which SQL statement adds a new record to a table?

- a) ADD RECORD
- b) INSERT INTO
- c) CREATE ROW
- d) APPEND

78. What does the JOIN clause do in SQL?

- a) Combines rows from two or more tables
- b) Joins two databases
- c) Merges columns within a table

- d) Creates a new table

79. Which function returns the number of rows in a result set?

- a) COUNT()
- b) SUM()
- c) TOTAL()
- d) ROWS()

80. What is a FOREIGN KEY used for?

- a) To encrypt external data
- b) To link two tables together
- c) To create a backup
- d) To validate input

81. Which SQL statement updates existing data?

- a) MODIFY
- b) CHANGE
- c) UPDATE
- d) ALTER

82. What does the DISTINCT keyword do?

- a) Sorts results
- b) Returns only different (unique) values
- c) Filters null values
- d) Groups results

83. Which clause is used with aggregate functions to group results?

- a) ORDER BY
- b) GROUP BY
- c) FILTER BY
- d) SORT BY

84. What is the correct syntax to delete a record?

- a) DELETE FROM table WHERE condition;
- b) REMOVE FROM table WHERE condition;
- c) DROP record FROM table;
- d) ERASE FROM table;

85. Which SQL data type stores dates?

- a) STRING
- b) INTEGER

- c) DATE
- d) TIMESTAMP (also valid for date+time)

86. What does the LIKE operator do?

- a) Compares exact values
- b) Searches for a specified pattern
- c) Checks for NULL
- d) Joins tables

87. Which statement creates a new table?

- a) MAKE TABLE
- b) NEW TABLE
- c) CREATE TABLE
- d) ADD TABLE

88. What is normalization in databases?

- a) Speeding up queries
- b) Organizing data to reduce redundancy
- c) Encrypting sensitive data
- d) Backing up data

89. Which command commits a transaction in SQL?

- a) SAVE
- b) COMMIT
- c) END
- d) FINISH

90. What does SELECT * FROM users do?

- a) Deletes all users
- b) Selects all columns from the users table
- c) Creates a new users table
- d) Updates all users

Security & Best Practices (Questions 91-100)

91. What is SQL injection?

- a) A database optimization technique
- b) A code injection technique that exploits SQL vulnerabilities
- c) A method to import SQL files
- d) A type of database backup

92. Which practice helps prevent SQL injection in PHP?

- a) Using `mysql_query()`
- b) Using prepared statements
- c) Escaping output only
- d) Hiding database credentials

93. What is XSS (Cross-Site Scripting)?

- a) A database attack
- b) Injecting malicious scripts into web pages viewed by users
- c) A server overload technique
- d) A CSS vulnerability

94. Which function helps prevent XSS in PHP output?

- a) `addslashes()`
- b) `htmlspecialchars()`
- c) `strip_whitespace()`
- d) `encode_url()`

95. What is the principle of "separation of concerns" in web development?

- a) Using multiple servers
- b) Keeping HTML, CSS, JS, and PHP in separate files/layers
- c) Separating development and production environments
- d) Using different databases for different features

96. Why should you validate input on both client AND server side?

- a) Client-side is faster only
- b) Client-side can be bypassed; server-side is essential for security
- c) Server-side validation is optional
- d) To reduce server load only

97. What is the purpose of using HTTPS?

- a) Faster page loading
- b) Encrypting data transmitted between client and server
- c) Better SEO ranking
- d) Compressing images

98. Which practice is recommended for storing passwords?

- a) Plain text
- b) MD5 hash
- c) `password_hash()` with `bcrypt/argon2`

- d) Base64 encoding

99. What does "CSRF" stand for?

- a) Cross-Site Request Forgery
- b) Client-Side Resource Failure
- c) Cached Server Response Filter
- d) Code Security Risk Factor

100. Why use version control (e.g., Git) in web projects?

- a) To compress files
- b) To track changes, collaborate, and revert if needed
- c) To automatically deploy code

PART 2: Data Manipulation – SQL

Consider the relational schema of a **vehicle rental management database**:

- **Agence**(AgenceID, Name, City, Telephone)
- **Vehicule**(VehiculeID, Brand, Model, Category, #AgenceID, Available)
- **Client**(ClientID, Name, FirstName, LicenseNumber, Telephone)
- **Location**(LocationID, #ClientID, #VehiculeID, StartDate, EndDate, MileageStart, MileageEnd, Amount)

Primary keys are underlined and foreign keys start with #.

Write SQL queries for the following:

1. Create the **Location** table with all necessary integrity constraints

(primary keys, foreign keys, and domain constraints).

2. List the vehicles (**brand, model**) that are currently **unavailable** at the agency located in **Douala**.

3. Display, for each client, the **total number of rentals** made.

4. Find the clients who have rented **at least one vehicle** in the **“4x4”** category.

5. Compute the **average mileage traveled per rental** for each vehicle brand.

PART 5: Software Engineering

1. Define: Software engineering, requirements specification (software requirements document), software, prototype, life cycle.

2. Give four qualities of a good requirements specification with explanations.

3. State two advantages and two disadvantages of the waterfall model.

4. Why is the use of a methodology essential in software engineering?